

PRODUCTION LIFECYCLE MANAGEMENT SYSTEM

Setup & User

Guide

Complete installation, configuration, and expected outcomes for the FL Studio Production Tracker.



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SECTION 1 — SYSTEM OVERVIEW

What is PLMS?

The Production Lifecycle Management System (PLMS) is a full-stack web application built in Python and Flask for Kweza Arts. It wraps around FL Studio's workflow to provide structured version tracking, quality scoring, change analysis, and predictive insight across every stage of a music production project — all accessible through a professional, browser-based dashboard.

V

Version Capture

Auto

Automatically snapshots every FL Studio save with full metadata and timestamps.

C

Change Engine

Python

Computes measurable differences in tempo, channels, and patterns between versions.

S

Quality Scoring

ML

Assigns a 0–100 score to each version based on configurable production metrics.

P

Prediction

sklearn

Uses linear regression to forecast how your track will evolve across future versions.

T

Timeline View

Live

Horizontal scrollable DAW-inspired version timeline showing every saved state.

R

Rollback

Safe

Safely restore any previous version of a project with one click.

Technology Stack

Component	Technology	Purpose
Web Framework	Flask 3.0	Routing, templates, session management
Database	SQLite + SQLAlchemy	Persistent storage for projects and versions
Authentication	Flask-Login + Werkzeug	Secure user sessions and password hashing
FLP Parsing	pyflp	Reads FL Studio project file metadata
File Watcher	watchdog	Detects .flp saves in real time
Analysis	NumPy + scikit-learn	Change computation and predictions
Frontend	Jinja2 + Chart.js	Responsive dashboard with live charts
Icons	Lucide Icons	Professional SVG icon system

SECTION 2 — INSTALLATION & SETUP

System Requirements

Requirement	Minimum	Recommended
Python	3.10+	3.12
OS	Windows 10 / macOS 12 / Ubuntu 22	Windows 11
RAM	2 GB	4 GB
Disk	200 MB	1 GB (for backups)
Browser	Chrome 110 / Firefox 110	Chrome latest
FL Studio	20+ (optional)	21 (for live watcher)

Step-by-Step Installation

1

Download and Extract

Download plms_app.zip from the provided link.

```
unzip plms_app.zip
```

```
cd plms
```

You should see: app.py, models.py, engines.py, watcher.py, requirements.txt, templates/, static/

2

Create a Python Virtual Environment

Open a terminal in the plms/ folder and run:

```
python -m venv venv
```

Activate it:

```
Windows: venv\Scripts\activate
```

```
Mac/Linux: source venv/bin/activate
```

Your terminal prompt should now show (venv).

3

Install Dependencies

With the venv active, install all required packages:

```
pip install -r requirements.txt
```

This installs Flask, SQLAlchemy, Flask-Login, watchdog, pyflp, NumPy, and scikit-learn.

4

Run the Application

Start the Flask development server:

```
python app.py
```

Expected output:

```
* Running on http://127.0.0.1:5000
```

Open your browser and go to <http://localhost:5000>

Note: The first run automatically creates a `plms.db` SQLite database in the project folder. No manual database setup is required.

SECTION 3 — FIRST USE & AUTHENTICATION

Creating Your Account

PLMS uses a secure authentication system. Every user has a private workspace — projects, versions, and settings are never shared between accounts.

1

Open the Registration Page

Navigate to `http://localhost:5000` in your browser.

You will be redirected to the login page.

Click "Create one" to go to the register page.

2

Fill in Your Details

Username: at least 3 characters (e.g. mzati)

Email: a valid email address

Password: at least 6 characters

Confirm Password: must match exactly

3

Sign In

After registering, you are automatically signed in.

On future visits, use your username or email + password.

Check "Keep me signed in" to stay logged in for 30 days.

Loading Demo Data

If you want to explore the system immediately without an FL Studio installation, PLMS includes a one-click demo data loader that creates three realistic projects with full version histories. To use it:

- Navigate to Settings (sidebar, bottom).
- Scroll to the Database section.
- Click "Load Demo Projects".
- Return to the Dashboard — three projects will appear: Summer Anthem (Afrobeats), Midnight Groove (Amapiano), and Afro Fusion Vol.1.

SECTION 4 — USING THE APPLICATION

Application Layout

PLMS uses a three-zone layout visible on all pages after login:

Zone	Location	Contents
Sidebar	Fixed left (260px)	Brand, navigation, project list, user profile
Top Bar	Fixed top	Page title, context actions, watcher status badge
Main Area	Centre/right, scrollable	Page-specific content: dashboard, timeline, charts

Dashboard

The Dashboard is the home screen. It provides a global view of your studio:

- Four stat cards at the top: Total Projects, Total Versions, Average Quality Score, Watcher Status.
- A responsive grid of project cards, each showing name, genre, version count, BPM, channel count, quality score, and trend badge.
- A "New Project" button (top right and sidebar) to create a project at any time.
- An empty state with a "Load Demo Data" shortcut if no projects exist yet.

Creating a Project

- Click "New Project" in the top bar or the sidebar "+" button.
- Enter a project name (required), genre, FL Studio file path, and optional description.
- Click "Create Project" — you are taken directly to the project detail page.
- The file path tells the watcher which .flp file to monitor for automatic captures.

Project Detail Page

The Project Detail page is the core of PLMS. It contains all analytical information for a single project:

Component	Description
Stat Row	Latest score, total versions, current BPM, channel count
Version Timeline	Horizontal scrollable cards — newest first — showing score, BPM, channels, patterns, and date. "LATEST" badge
Latest Changes	Diff between last two versions: what increased, what decreased, by how much.
Predictions	Linear regression forecast for next version: expected score, channel count, tempo, and trend direction.
Charts (4)	Quality score over time · Channel growth · Tempo evolution · Pattern complexity
Simulate Save	Button to add a demo version (realistic random progression) without FL Studio.
Rollback	Each non-latest version card has a Rollback button to restore that file state.

SECTION 5 — FL STUDIO LIVE WATCHER

Connecting FL Studio

For fully automatic version capture, configure PLMS to watch the folder where FL Studio saves your project files. Every time you press Ctrl+S inside FL Studio, PLMS detects the change and captures a new version snapshot within seconds.

1

Set the Watched Folder

Go to Settings > General.

Enter the full path to your FL Studio projects folder:

Windows: `C:\Users\YourName\Documents\FL Studio\Projects`

Mac: `/Users/YourName/Documents/FL Studio/Projects`

Click "Save General".

2

Create a Project with the .flp Path

Create or edit a project and set its File Path to the exact .flp file you want to track, e.g.:

`C:\...\Projects\SummerAnthem.flp`

PLMS matches incoming watcher events to projects by filepath.

3

Start the Watcher

Go to Settings > File Watcher.

Click "Start Watcher".

The badge in the top bar changes to "Watcher On" with a pulsing dot.

Save any .flp file in FL Studio — a new version appears automatically.

Without FL Studio: Use the "Simulate Save" button on any project page to add demo versions with realistic random progressions.

How Version Capture Works

When a .flp save is detected, PLMS performs the following steps automatically:

1	Detect	watchdog fires an on_modified event for the .flp file
2	Parse	pyflp reads tempo, channel count, and pattern count from the file
3	Score	ScoringEngine calculates a 0–100 quality score
4	Diff	ChangeEngine computes what changed vs. the previous version
5	Store	New Version row written to SQLite with all metadata

SECTION 6 — SETTINGS & CONFIGURATION

Settings Reference

Setting	Location	Default	Description
Watched Folder	General	./projects	Folder monitored for .flp saves
Notifications	General	On	Desktop alert on new version capture
Auto Backup	General	On	Copy .flp to backups/ on capture
Watcher	Watcher	Off	Toggle real-time file monitoring
Weight: Tempo	Scoring	25	Score contribution from BPM range
Weight: Channels	Scoring	35	Score contribution from track richness
Weight: Patterns	Scoring	25	Score contribution from pattern complexity
Weight: Maturity	Scoring	15	Score contribution from version count
Password	Account	—	Change login password
Load Demo Projects	Database	—	Seed 3 demo projects with version history

Scoring Weight Configuration

The scoring engine uses four configurable weights. They do not need to sum to 100 — each dimension is independently normalised before being multiplied by its weight. Adjust weights based on your genre priorities:

- Afrobeats / Amapiano: increase Channels weight (layered percussion is key).
- Minimal / Lo-fi: reduce Channels, increase Maturity (refinement matters more).
- EDM / Dance: increase Tempo weight (BPM precision is critical).
- Experimental: reduce all structural weights, increase Maturity.

SECTION 7 — EXPECTED OUTCOMES

What to Expect from PLMS

The following section details the expected behaviours, outputs, and quality indicators at each stage of using PLMS. Use this as a reference to verify correct operation and to evaluate your production progress.

Registration & Login

● Successful registration	Redirected to Dashboard with "Account created" toast
● Duplicate username	"Username already taken" error shown on register page
● Wrong password on login	"Invalid username or password" error — no account details revealed
● Remember me enabled	Session persists for 30 days without re-login
● Protected page without login	Automatically redirected to /login

Project Creation & Management

● New project created	Redirected to project detail page immediately
● No versions yet	Empty timeline shown with "Simulate Save" prompt
● First version added	Timeline shows 1 card with "LATEST" badge, score calculated
● Second version added	Changes panel shows diffs, score bar animates
● Project deleted	"Project deleted" toast, project removed from sidebar list
● Demo data loaded	3 projects appear: 6, 4, and 7 versions respectively

Version Capture & Scoring

Quality scores follow a predictable pattern as projects mature. Below are typical score ranges by production stage:

Stage	Typical Score	Interpretation
Initial sketch (v1–v2)	20 – 40	Few channels, minimal patterns — expected at start
Early development (v3–v5)	40 – 60	Growing arrangement, tempo settled
Mid production (v5–v8)	55 – 75	Full arrangement, active mixing
Late stage (v8+)	70 – 90	Mature, complex, many refined versions
Fully produced (v10+)	80 – 100	All metrics near maximum, high maturity bonus

Prediction Engine

 Fewer than 3 versions	"Need at least 3 versions to predict" message shown
 3–4 versions available	Predictions shown with "Moderate confidence" badge
 5+ versions available	Predictions shown with "High confidence" badge
 Upward score trend	Trend label: "Improving" or "Growing" in green
 Flat score across versions	Trend label: "Stable" in grey
 Declining score	Trend label: "Declining" in red — review latest changes

File Watcher

 Watcher started	"Watcher On" badge with pulsing green dot in top bar
 .flp saved in watched folder	New version appears on project detail within 2–3 seconds
 File saved rapidly (debounce)	Only one version created per 2-second window
 Watched folder not found	Folder is created automatically on watcher start
 pyflp unavailable	Capture falls back to default metadata (BPM 120, 0 channels)

SECTION 8 — TROUBLESHOOTING

Common Issues

Port 5000 already in use

Cause Another process is using port 5000.

Fix Run: `python app.py` and edit the last line to use `port=5001`, or kill the process using port 5000.

pyflp import error

Cause `pyflp` is not installed or incompatible with your FL Studio version.

Fix Run: `pip install pyflp`. If it still fails, PLMS will use default metadata — the app will still work.

Watcher not detecting saves

Cause The watched folder path is wrong, or FL Studio saves to a different location.

Fix In Settings, verify the watched folder matches exactly where FL Studio saves your files. Check for trailing slashes.

Charts not loading

Cause The project has fewer than 2 versions, or Chart.js CDN is unreachable.

Fix Add at least 2 versions. For offline use, download Chart.js locally and update the `base.html` script tag.

Login redirect loop

Cause Browser is blocking cookies or the `SECRET_KEY` changed.

Fix Clear browser cookies for `localhost:5000`, then sign in again.

Database locked error

Cause Multiple Flask processes are running simultaneously.

Fix Stop all running instances with `Ctrl+C`, then restart: `python app.py` with `use_reloader=False` (already set).

Quick Command Reference

Install dependencies	<code>pip install -r requirements.txt</code>
Start application	<code>python app.py</code>
Access in browser	<code>http://localhost:5000</code>
Stop application	<code>Ctrl + C</code> in terminal
Deactivate venv	<code>deactivate</code>

For academic or technical queries, contact the project supervisor Mrs Funsani or refer to the full project proposal document.